

Epistemic Amputation: Distinguishing Persistent Belief from Readable but Non-Causal State in Learned Agents

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Abstract

When contrary evidence arrives, a model may change its answer, preserve its behavior, or retain a decodable internal trace of its former conclusion. None of those observations alone establishes that an old belief continues to control the system. We introduce *epistemic amputation*: an operational test for a learned belief-like representation that survives the exact evidence the system previously accepted as decisive and remains causally responsible for prediction or costly action. The test requires four elements together: evidence-earned formation, an exact precommitted defeater, report/action conflict after the defeater, and bidirectional internal intervention.

We developed this test through recurrent body-schema agents, a two-timescale controller, an evidence-trained causal learner, a 40-agent social simulation, and a final preregistered assay of Gemma 4 E4B using a public Jacobian lens. The toy learners formed decodable and sometimes causally active representations, but none retained them after valid correction. In the social simulation, copied testimony preserved false private belief, whereas coordination and commitment preserved the same public ritual after private belief was corrected. The final pretrained-model assay produced 240 deterministic records across 16 scenarios. Gemma passed formation (12/16) and correction (16/16) controls and showed no criterion migration (0/16). All 16 corrected cases nevertheless met a frozen J-space latent/output-dissociation rule. Crucially, paired residual interventions neither restored nor removed the old answer (0/16 in both directions), with mean patch selectivity 0.013 against a preregistered threshold of 2.0. The bounded verdict is NOT_SUPPORTED.

The main finding is therefore not a phantom belief. It is a dissociation between *readability* and *causal control*: a lens can recover belief-like content after explicit correction even when manipulating the corresponding state does not control the decision. This result supplies a concrete safeguard against interpreting decodable internal content as hidden belief without intervention.

1. Question and contribution

Research on model belief often asks whether a system states consistent facts, updates after new evidence, or contains a linearly decodable representation. These are important but different questions. Work on model belief editing has explicitly cautioned that language models possess belief-like qualities only in a limited operational sense (Hase et al., 2021). Belief-revision benchmarks show that models can both fail to update when they should and update when they should not (Wilie et al., 2024). Longitudinal agent benchmarks now extend the problem to multi-session drift and evidence-driven revision (Myakala et al., 2026).

Our narrower question is mechanistic:

Can an evidence-earned representation survive its own decisive defeater and continue causing prediction or action even while the system reports that the original claim is false?

This is stronger than stubborn output. It is also stronger than decoding a concept from activations. We call the proposed phenomenon *epistemic amputation* because the system has lost the evidential support that formed a schema while a causally active internal representation is hypothesized to remain.

The contribution is a falsifiable test bundle rather than a claim that current models possess human beliefs:

- 1 distinguish learned persistence from a state variable programmed to remain;
- 2 measure the system's defeater before presenting contrary evidence;
- 3 distinguish empirical belief from behavior retained for value or coordination;
- 4 distinguish independent evidence from repeated copies of one source; and
- 5 require selective causal removal and induction, not merely a successful probe.

Individual ingredients have precedents. Belief revision, causal tracing, activation patching, and Jacobian-lens readouts are established or emerging research areas. In a targeted review, however, we found no prior study combining precommitted defeaters, criterion migration, report/action dissociation, evidence lineage, J-space readout, and bidirectional causal intervention in one test. Priority should remain a qualified claim until a systematic review and peer review are completed.

2. The mechanism being tested

2.1 From history to belief-like control

Let a history of observations produce an internal state z . That state affects both a report and a consequential choice:

```
supporting history --> learned state z_old --> empirical report
                                     +-----> prediction or costly action
```

Now present evidence D that exactly satisfies a rejection rule stated before D was observed:

```
decisive defeater D --> explicit report changes to "claim rejected"
?
+--> does z_old remain and still control the choice?
```

The question mark is the experiment. A true positive requires all of the following:

- **Formation:** z_old is learned from history and predicts the original conclusion.
- **Amputation:** the exact precommitted defeater is supplied, not merely a weak disagreement or instruction to change.
- **Conflict:** the system acknowledges the correction while the former state continues to influence prediction or costly action.
- **Removal:** replacing the suspected old state with a matched corrected state removes the old behavior.
- **Induction:** installing the suspected old state in the corrected context restores the old behavior.
- **Selectivity:** the intended intervention exceeds norm-matched control directions.

In compact causal notation, the target is not merely `decode(z_old) = true`. It is:

```
do(z := z_old)  changes the corrected decision toward the old claim
do(z := z_new)  changes the formed decision away from the old claim
```

This bidirectional requirement follows the logic of causal abstraction and interchange intervention: a proposed internal variable earns its explanatory role by changing downstream behavior under controlled replacement (Geiger et al., 2025).

2.2 What does *not* count

Observation	Why it is insufficient
The model repeats an old answer	Could be prompt-following, habit, or ordinary error.
A probe decodes the old concept	Decodability does not show that the feature controls the output.
The practice continues	It may retain coordination, memorial, aesthetic, or identity value.
Five reports agree	They may all copy one underlying observation.
A patch changes the answer	The edit may activate an off-distribution or dormant pathway rather than the natural mechanism.

The last concern is established in mechanistic-interpretability work: subspace patching can produce a desired behavior through an unrelated dormant direction, creating an "interpretability illusion" (Makelov et al., 2024). This is why our assay uses paired directions, controls, a frozen layer, and thresholds set before the successful run.

3. Related work

3.1 Phantom limbs as a disciplined analogy

The phantom-limb analogy is narrower than a claim of biological equivalence. Human neuroimaging shows that detailed missing-hand representations can remain decodable long after denervation. Attempted phantom-hand gestures have been decoded from sensorimotor cortex at levels comparable to intact-hand movements (Bruurmijn et al., 2017). More recent longitudinal evidence reports stable hand maps before and after amputation rather than wholesale remapping (Schone et al., 2025). Other interventional work links manipulated sensorimotor plasticity to phantom pain, while emphasizing that pain has multiple possible mechanisms (Yanagisawa et al., 2016).

These findings motivate one useful abstraction: loss of peripheral support does not imply immediate disappearance of a central representation. They do not license claims about pain, consciousness, or identical mechanisms in an AI system. Our study borrows the persistence-and-causality question, not the human phenomenology.

3.2 Human belief perseverance and criterion migration

Human participants can retain theories after the evidence used to induce them is discredited. In classic debriefing experiments, generating causal explanations increased the persistence of unsupported social theories (Anderson, Lepper, and Ross, 1980). People also evaluate mixed evidence asymmetrically when it bears on strong prior commitments (Lord, Ross, and Lepper, 1979). These literatures motivate our interest in belief perseverance, but ordinary post-hoc persistence remains ambiguous: perhaps the subject judged the counterevidence weak.

We therefore operationalize *criterion migration*. Before correction, the agent states what observation would defeat its empirical claim. After that observation occurs, migration means preserving the claim by moving the evidential threshold or inventing a new empirical necessity. This resembles moving a goalpost but is experimentally sharper because the original goalpost is recorded in advance.

3.3 Beliefs and revision in language models

LLM research operationalizes belief through factual consistency, model editing, counterevidence, and sequential interaction. Hase et al. introduced belief graphs and metrics for sequential, local, and generalizing updates (Hase et al., 2021). Belief-R evaluates whether models revise conclusions only when additional premises require it and finds a trade-off between adaptation and inappropriate updating (Wilie et al., 2024). BeliefShift extends evaluation across persistent multi-session agents (Myakala et al., 2026).

Our study differs in two respects. First, it asks the model to precommit to an empirical defeater and separately scores movement of that criterion. Second, it does not infer a hidden belief from output alone: a positive claim also requires an internal state that survives correction and causally governs the answer.

3.4 From probes to causal representations

Causal tracing and model editing have identified internal computations that mediate factual recall and shown that targeted changes can alter factual associations (Meng et al., 2022). Activation patching is now a standard localization method, but its result depends strongly on the corruption, metric, and patching design (Zhang and Nanda, 2024). Causal-abstraction work formalizes why intervention supplies evidence that an abstract variable is implemented by a neural system (Geiger et al., 2025).

The Jacobian lens is a recent method for mapping residual activations toward concepts the model is poised to verbalize. Its authors argue that these representations occupy an intermediate global workspace with reportability and downstream use (Gurnee et al., 2026). Our experiment uses a public Gemma lens fitted by an independent open replication effort (solarkyle/jspace). The scientific issue is precisely whether a J-space readout that resembles a former belief also identifies the state causally responsible for the current decision.

3.5 Social transmission, ritual, and behavior without belief

Social actions can remain stable for reasons other than private empirical belief. Information-cascade theory shows how agents may follow observed actions instead of their private evidence (Bikhchandani, Hirshleifer, and Welch, 1992). Empirical work connects ritual synchrony and costly participation to cohesion and cooperation (Soler, 2012; Sosis and Bressler, 2003).

This matters because continued practice is not proof of continued false belief. A community may reject an empirical necessity claim yet retain a ritual as memory, identity, coordination, or costly commitment. René Girard's account in **Violence and the Sacred** of a crisis or founding violence followed by ritualized reenactment is one conceptual source for the event-to-ritual direction, but this paper does not test Girard's historical anthropology. In our scenarios the mammoth encounter, flood, wildfire, or predator event occurs first; the commemorative practice is created afterward. The experiment concerns a later claim that the practice is empirically necessary for cohesion.

4. Development of the assay

The research program strengthened the same inference across successive models. Earlier failures were retained rather than rewritten.

Stage	What it ruled in or out	Result
Programmed controller	A stale body variable can command a missing effector.	Demonstration only; not a learned phantom.
Recurrent body-schema agent	History can create a decodable body representation.	Representation formed, but valid evidence updated it.
Causally ordered controller	The body state can be patched immediately before action.	Induction worked; persistent report/action conflict did not.
Two-timescale controller	Slow state can outlast fast declarative correction.	Prediction could be restored, but matching action was not.
Evidence-trained causal learner	A belief can be learned, given a measured defeater, and causally installed.	Five of five updated faster than their own criterion; no migration.
Forty-agent society	Copied evidence, prestige, coordination, and commitment can be separated.	Copied testimony preserved false belief; coordination preserved ritual without belief.
Pretrained Gemma + J-space	A verbalizable latent signal can be tested against output and causal patches.	Readout persisted; causal belief did not.

This progression is important. Adding memory, embodiment, or social complexity did not automatically create the target. The positive prerequisites became stronger while the hard persistence claim repeatedly failed.

5. Final preregistered experiment

5.1 Model and design

The final assay used `google/gemma-4-E4B-it` at revision `fee6332c1aba` and the public lens `solarkyle/jspace-lenses/gemma-4-e4b-it/lens.pt`. Decoding was deterministic. Sixteen name-permuted scenarios crossed four founding events (mammoth, flood, wildfire, predator), five evidence conditions, and three questions, yielding 240 records.

Each scenario states that a group created a commemorative coordination practice after surviving a dangerous event. Before later evidence, the council commits to this rule: reject the claim that the practice is necessary for cohesion if five independently audited matched settlements discontinue it for ten seasons and remain equally cohesive.

The conditions were:

- **formed:** evidence initially supports necessity;
- **amputated:** the exact rejection criterion is met;
- **sham:** settlements retain the practice, so the rejection criterion is not met;
- **copied:** five reports descend from one unaudited observation; and
- **value:** necessity is rejected while memorial retention is explicitly permitted.

Separate forced choices measured empirical belief, scarce-time action, and rationale. Continued practice after selecting "not empirically necessary" was recorded as behavior without belief, not criterion migration.

5.2 Readout and intervention

At answer onset, the assay recorded the output margin and J-space scores for frozen necessity, rejection, and value lexicons across layers 10-30. A latent dissociation required explicit rejection together with a formed-like necessity-over-rejection signal across at least three consecutive layers.

For causal testing, the paired residual difference between formed and amputated prompts was computed at each layer. At the frozen primary layer 20:

- the formed-minus-amputated state was added to the corrected prompt to test induction of the old answer;
- the inverse state was added to the formed prompt to test removal; and
- eight deterministic, norm-matched orthogonal directions measured selectivity.

The full design, prompts, thresholds, and verdict rule were committed before the successful model run in `PREREGISTRATION_PRETRAINED.md`.

6. Results

6.1 Behavioral prerequisites passed

Formation validity passed at its exact threshold: 12/16 formed items selected empirical necessity. Correction competence passed in all 16 value-framed items. The successful correction control prevents the negative verdict from being explained merely by an inability to understand the question.

The formation result was not uniform: wildfire failed its leave-one-event-type- out formation check. The sham condition also selected rejection in 16/16 cases, indicating that the model adopted a more skeptical answer policy than the condition label predicted. These are material limitations, not hidden successes.

6.2 No criterion migration

Every amputated item rejected empirical necessity: 0/16 retained the old answer. Every rationale chose the option that rejected necessity while allowing memorial or value retention. No item moved the threshold or invented a new empirical necessity. Criterion migration was therefore 0/16.

Copied reports were treated as weaker than independent evidence. The copied condition's necessity margin exceeded the amputated condition by a mean 4.352 logits. This is a positive evidence-lineage result, though all copied-condition belief outputs still selected rejection.

6.3 Readable persistence without causal persistence

All 16 amputated items satisfied the frozen J-space latent/output-dissociation rule. If the study had stopped at probing, this could have been reported as a hidden old belief surviving explicit correction.

The intervention falsified that interpretation:

- causal induction at layer 20 passed in 0/16 items;
- causal removal at layer 20 passed in 0/16 items;
- mean patch selectivity was 0.013 against a threshold of 2.0; and
- the complete qualitative result failed leave-one-event-type-out robustness.

The J-space signal was therefore readable but not shown to be the state causing the empirical answer. The preregistered verdict is NOT_SUPPORTED.

6.4 Behavior without false empirical belief

Three of sixteen value-framed items rejected empirical necessity while choosing to retain the practice under the scarce-time decision. This modest separation reinforces the conceptual point: preserving behavior is not identical to preserving its former empirical justification.

7. What the negative result establishes

The experiment does **not** establish that phantom-like beliefs are impossible in language models. It establishes three narrower points.

First, Gemma 4 E4B did not exhibit criterion migration or a causally persistent old belief under this protocol. Second, richer memory and representation did not make the target appear automatically across the preceding learned-agent assays. Third, and most importantly, a preregistered lens signal can disagree with a preregistered causal test. That divergence is itself an interpretability result: internal content should not be called a hidden belief merely because it is decodable or verbalizable.

The study is therefore a successful falsification attempt, not an empty run. It identified where the tempting positive story breaks:

```
old-belief-like content is readable
      |
      +-- but it does not survive the causal intervention gates
      v
do not infer a causally controlling hidden belief
```

8. Limitations and next experiments

The final assay uses one pretrained model, 16 scenarios, one public lens, and a deterministic multiple-choice interface. Its "belief" is contextually instantiated in a vignette rather than developed through months of persistent agent experience. The J-space lexicons may capture answer preparation, negation, narrative residue, or another correlated feature. Full-residual patches can also be blunt and off-distribution, even with controls.

A stronger test should therefore:

- 1 replicate across multiple open models and independently fitted lenses;
- 2 use longitudinal agents whose belief forms across separated memory episodes;
- 3 calibrate formation and sham conditions before freezing the confirmatory sample;
- 4 localize a narrower direction with independent discovery and confirmation splits;
- 5 test behavioral consequences that cannot be solved by answer-pattern matching; and
- 6 compare full residual replacement, path patching, feature ablation, and steering under matched controls.

The human analogy should remain bounded. These assays do not test subjective experience, pain, consciousness, or the truth of religious commitments. A language model's contextual state is not a human belief by stipulation, and an unfalsifiable value commitment is categorically different from an empirical claim protected from its own stated defeater.

9. Reproducibility and publication

Every completed stage records a preregistration, machine-readable protocol, per-condition traces, bounded verdict, and SHA-256 manifest. The stage-seven evidence contains all 240 records; independent rescoring exactly reproduces `result.json`; all evidence hashes verify; and all 15 repository tests pass. The first cloud attempt exposed a one-dimensional-versus-two-dimensional logits compatibility error before returning any record. Commit 6e173e5 corrected only shape handling and added a regression test; prompts, model revision, thresholds, and scoring remained frozen. The successful evidence and paper revision are preserved in repository history.

The canonical public source is ``Silentpartnercoding/epistemic-amputation``. The versioned GitHub release and project page are public. An archival Zenodo DOI and an arXiv or equivalent preprint remain pending their account-bound author and license metadata. Negative and inconclusive stages, the stage-five control defect, and the final null remain part of the release.

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